

Minimal Fibre Structure of the Non-Injective
Projection
from Born–Infeld Indiscernability:
Derivation of the Parity Involution
O18 — Spectral Admissibility Sub-programme

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Abstract

The identification of conjugate Weil blocks $\{c, q - c\}$ as the fibres of the non-injective projection $\Pi : \chi \rightarrow \mathcal{O}$ was stated as a structurally motivated hypothesis in O16 [22] and used in O17 [23] to derive the pair-level observable and the exponent doubling $\delta_{\text{pair}} = 2\delta_c \approx 7.44$. The present paper provides the foundational derivation of the underlying involution. We proceed in two stages. At the abstract level, we show that the Born–Infeld action $S[\chi]$ is even in χ , and we define a notion of *BI-indiscernability* for configurations that produce identical effective responses under all BI-admissible perturbations. Evenness of S immediately implies that χ and $-\chi$ are BI-indiscernible, so every projective fibre contains the orbit $\{\chi, -\chi\}$. We then prove a conditional minimality result: in the absence of any symmetry of $S[\chi]$ beyond its parity, the minimal fibre is exactly the involution $\chi \mapsto -\chi$. At the concrete level, the O17 result $\rho_{q-c} = \overline{\rho_c}$ together with the norm-invariance of Gram–Schmidt orthogonalisation identifies the involution $c \leftrightarrow q - c$ as the Weil-representation instance of the abstract parity. This closes the logical gap in O16–O17 by deriving the fibre structure from first principles rather than postulating it.

Keywords. fibre-level admissibility, Born–Infeld parity, BI-indiscernability, parity involution, non-injective projection, Weil representation, Heisenberg Cayley graphs, pair observable, observable hierarchy

Contents

1	Context and scope	3
2	Abstract level: parity from Born–Infeld	3
2.1	Imported results on non-injectivity	3
2.2	Evenness of the Born–Infeld action	4
2.3	BI-indiscernability and the fibre constraint	4
2.4	Minimality of the involution (conditional result)	5
3	Concrete level: Weil realisation in $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$	6
3.1	The setting	6
3.2	Imported result from O17	6
3.3	Identification of the abstract involution	6
4	Implications and relation to O16–O17	7
4.1	Requalification of O16 and O17	7
4.2	Effect on $\delta_{\text{pair}} \rightarrow \beta^*$	7
4.3	Open directions	8
5	Summary	9

1 Context and scope

The Cosmochrony framework [1] describes physics as emerging from a pre-geometric substrate field χ projected onto effective observables via a map $\Pi : \chi \rightarrow \mathcal{O}$. A central structural result of the framework is that Π must be non-injective: injectivity would imply trivial holonomy and therefore preclude any emergent gauge structure [2] (Born–Infeld companion paper, Proposition 1 and Corollary 1). Finite projective capacity, a consequence of non-injectivity, selects the Born–Infeld completion of the effective action [2].

The O-series spectral admissibility sub-programme has investigated the capacity exponent δ governing projective dynamics on Heisenberg Cayley graphs $\text{Cay}(G_q, S_q)$ with $G_q = \text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ for prime q . Papers O12–O15 [17, 18, 19, 21] established the exact block-level exponent $\hat{\delta}_{\text{exact}} \approx 3.72$, structurally below the phenomenological target $\delta \in [7.4, 10.6]$. O16 [22] identified the resolution: the physically relevant observable lives at the level of conjugate pairs $\{c, q - c\}$, yielding $\delta_{\text{pair}} = 2\delta_c \approx 7.44$. O17 [23] provided the structural derivation showing that the Gram–Schmidt dynamics of blocks c and $q - c$ are identical, establishing $\delta_{q-c} = \delta_c$ and that the factor $r(c, q)$ in O16 is a normalisation artefact, not an invariant.

However, O17 Section 5.2 explicitly identifies as the primary open problem of the fibre-level programme the derivation of the identification $c \leftrightarrow q - c$ as a necessary consequence of the projection structure Π and the Born–Infeld constraint, rather than a structurally motivated hypothesis. The present paper addresses this open problem.

Scope and what this paper does not claim. O18 establishes three things: (i) the Born–Infeld action is even in χ ; (ii) this evenness forces the projective fibres to contain the involution $\chi \mapsto -\chi$; (iii) in the Weil realisation of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, this involution is concretely $c \leftrightarrow q - c$. O18 does *not* claim that the involution is the unique possible fibre structure in all enriched frameworks: the minimality result is explicitly conditional on the absence of additional symmetries beyond Born–Infeld parity (Section 2). A general classification of all possible fibres is deferred to a subsequent paper.

2 Abstract level: parity from Born–Infeld

2.1 Imported results on non-injectivity

We recall two results established in the companion papers that are used here without re-derivation.

1. **Non-injectivity of Π [2].** If Π is injective and compatible with relational locality, the induced holonomy is trivial, precluding emergent gauge structure. Since electromagnetism must arise as an effective $U(1)$

fibre symmetry, Π must be non-injective. Non-trivial fibres $\Pi^{-1}(y)$ are therefore a structural requirement, not an auxiliary assumption.

2. **Born–Infeld selection** [2]. Non-injectivity implies finite projective transport capacity. Finite capacity, together with locality, Lorentz compatibility, and analyticity in the weak-field limit, selects the Born–Infeld effective action as the unique analytic completion ensuring saturation of the constitutive relation. The effective action takes the form

$$S_{\text{BI}}[\chi] = \beta^2 \int d^4x \left(1 - \sqrt{1 - \frac{F^2}{\beta^2}} \right), \quad F = D\chi, \quad (1)$$

where β is the saturation scale and $F^2 = F_{\mu\nu}F^{\mu\nu}$.

2.2 Evenness of the Born–Infeld action

Lemma 2.1 (Parity of S_{BI}). $S_{\text{BI}}[-\chi] = S_{\text{BI}}[\chi]$.

Proof. Under $\chi \mapsto -\chi$, the field gradient transforms as $F = D\chi \mapsto -D\chi = -F$. Therefore $F^2 = F_{\mu\nu}F^{\mu\nu} \mapsto (-F)_{\mu\nu}(-F)^{\mu\nu} = F^2$. Since the integrand in (1) depends on χ only through F^2 , the action is unchanged. $\square$

Remark 2.2. *Lemma 2.1 uses only the quadratic dependence on F in the argument of the square root, which is the defining property of the Born–Infeld structure. It does not depend on any discrete group structure and holds for all smooth χ .*

2.3 BI-indiscernability and the fibre constraint

Definition 2.3 (BI-indiscernability). *Two configurations χ_1, χ_2 are BI-indiscernible if, for every BI-admissible perturbation η (i.e. $\|D\eta\|$ bounded and $\|D(\chi_i + \epsilon\eta)\| \leq c_\chi$ for small $\epsilon > 0$), every effective response functional R constructed solely from S_{BI} and its derivatives — in particular, R depends on χ only through BI-admissible local invariants, hence only through $F^2 = (D\chi)^2$ and its covariant derivatives, with no residual orientation-sensitive dependence (i.e. responses invariant under BI-admissible perturbations at fixed resolution scale) — satisfies*

$$R[\chi_1 + \epsilon\eta] = R[\chi_2 + \epsilon\eta] \quad \text{for all admissible } \eta \text{ and all sufficiently small } \epsilon. \quad (2)$$

Corollary 2.4 (Involution is contained in every fibre). *For every χ , the pair $\{\chi, -\chi\}$ consists of BI-indiscernible configurations. Consequently, every projective fibre $\Pi^{-1}(y)$ contains the orbit $\{\chi, -\chi\}$ for each $\chi \in \Pi^{-1}(y)$.*

Proof. By Lemma 2.1, $S_{\text{BI}}[\chi] = S_{\text{BI}}[-\chi]$. Any response functional R built from S_{BI} and BI-admissible perturbations therefore satisfies $R[-\chi + \epsilon\eta] = R[\chi + \epsilon(-\eta)]$. The admissibility condition in Definition 2.3 requires $\|D\eta\|$ bounded and $\|D(\chi + \epsilon\eta)\| \leq c_\chi$; replacing η by $-\eta$ preserves both bounds, so the admissible set is closed under $\eta \mapsto -\eta$. Consequently, since $-\eta$ ranges over all admissible perturbations whenever η does, the response of χ under η equals the response of $-\chi$ under $-\eta$, and hence also under η . This yields $R[\chi + \epsilon\eta] = R[-\chi + \epsilon\eta]$ for all admissible η , so χ and $-\chi$ are BI-indiscernible by Definition 2.3. $\square$

2.4 Minimality of the involution (conditional result)

Corollary 2.4 shows that the parity orbit is *contained* in every fibre. We now address whether the parity orbit is the *minimal* fibre, i.e. whether any additional identification $\chi \sim \psi$ with $\psi \neq -\chi$ is forced by the Born–Infeld structure.

Theorem 2.5 (Conditional minimality of the parity fibre). *Suppose the only global effective symmetry of $S_{\text{BI}}[\chi]$ compatible with BI-admissible observables is the parity $\chi \mapsto -\chi$. Then the minimal BI-indiscernible fibre is the two-element orbit $\{\chi, -\chi\}$. Any identification $\chi \sim \psi$ with $\psi \neq \pm\chi$ either*

- (i) *implies the existence of a further effective symmetry of S_{BI} compatible with BI-admissible observables beyond parity, contradicting the hypothesis; or*
- (ii) *is unstable, in the sense that there exists a BI-admissible perturbation η for which $R[\chi + \epsilon\eta] \neq R[\psi + \epsilon\eta]$ for some response functional R .*

Proof. Suppose $\chi \sim \psi$ with $\psi \neq \pm\chi$, i.e. $R[\chi + \epsilon\eta] = R[\psi + \epsilon\eta]$ for all admissible η and all R . Setting $\epsilon = 0$, this requires $R[\chi] = R[\psi]$ for all R , hence $S_{\text{BI}}[\chi] = S_{\text{BI}}[\psi]$. The map $\chi \mapsto \psi$ therefore induces an invariance of all BI-derived observables, hence an effective symmetry of the action S_{BI} . By hypothesis, the only such effective symmetry is parity, so $\psi = -\chi$, a contradiction. If instead (2) holds at $\epsilon = 0$ but fails for some admissible η at positive ϵ , then case (ii) applies by definition. $\square$

Remark 2.6 (Status of the hypothesis). *The hypothesis that parity is the only global effective symmetry of S_{BI} compatible with BI-admissible observables is not proved here; it is a structural assumption about the effective action. Theorem 2.5 should therefore be read as: given the Born–Infeld structure alone, the minimal fibre is the parity orbit. If the effective action admits additional symmetries in an enriched framework (for instance, from additional topological or gauge constraints), the fibre could be larger. A classification of fibres in such enriched settings is deferred to a subsequent paper.*

The abstract-level conclusion is:

Theorem 2.7 (Abstract parity involution). *Under the Born–Infeld effective action (1) and in the absence of global effective symmetries of S_{BI} beyond parity, the projection Π has minimal fibres given by the involution*

$$\Pi(\chi) = \Pi(-\chi) \quad \text{for all } \chi \in \mathcal{C}_\chi, \quad (3)$$

i.e. Π factors through the $\mathbb{Z}_2$ -orbit $\{\chi, -\chi\}$.

3 Concrete level: Weil realisation in $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$

3.1 The setting

We work in the discrete Heisenberg group $G_q = \text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ for prime q , with the Cayley graph $\text{Cay}(G_q, S_q)$ where $S_q = \{X^{\pm 1}, Y^{\pm 1}\}$ and X, Y are the standard generators. The Weil representations ρ_c of G_q are parametrised by the character index $c \in (\mathbb{Z}/q\mathbb{Z})^\times$. The Gram–Schmidt redundancy $\sigma_c(n)$ at BFS shell n is the key spectral observable established in O12–O15 [17, 18, 19, 21].

3.2 Imported result from O17

The following result is Theorem 2.1 and Corollary 2.3 of O17 [23].

Theorem 3.1 (O17 Theorem 2.1 and Corollary 2.3). *The conjugation identity $\rho_{q-c} = \overline{\rho_c}$ together with the norm-invariance of the Gram–Schmidt process implies that the raw Gram–Schmidt redundancy counts are exactly identical for ρ_c and ρ_{q-c} when the initial supports coincide, and structurally isomorphic in general. In particular, $\delta_{q-c} = \delta_c$ for all admissible c .*

3.3 Identification of the abstract involution

Theorem 2.7 establishes that the Born–Infeld structure forces an involution $\Pi(\chi) = \Pi(-\chi)$. We now identify the concrete realisation of this involution in the Weil setting.

The conjugation $\rho_{q-c} = \overline{\rho_c}$ is a complex conjugation of representations. Complex conjugation is the natural realisation of a sign inversion at the level of representation amplitudes: if a representation ρ_c encodes the response of the substrate field χ at character c , then $\overline{\rho_c}$ encodes the response of $-\chi$ at the same character, since complex conjugation reverses the phase of the effective observable. The index shift $c \mapsto q - c$ is therefore the concrete group-theoretic realisation of the abstract sign inversion $\chi \mapsto -\chi$ within G_q .

This identification is consistent with, and independently supported by, Theorem 3.1: the dynamical indistinguishability of blocks c and $q - c$ under Gram–Schmidt is precisely the spectral manifestation of BI-indiscernability (Definition 2.3) in the Weil setting.

Remark 3.2 (Representation-level identification). *The correspondence $\chi \leftrightarrow c$, $-\chi \leftrightarrow q - c$ should be understood as a representation-level realisation of the abstract involution: the conjugation $\rho_{q-c} = \overline{\rho_c}$ is the concrete spectral instance of the sign inversion $\chi \mapsto -\chi$ within the Weil setting. A direct derivation of this correspondence from χ -space variables, bypassing the representation-theoretic language, is left for future work.*

Theorem 3.3 (Weil-level parity involution). *In the Weil realisation of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, the abstract parity involution $\Pi(\chi) = \Pi(-\chi)$ of Theorem 2.7 is concretely realised as*

$$\Pi(c) = \Pi(q - c) \quad \text{for all } c \in (\mathbb{Z}/q\mathbb{Z})^\times. \quad (4)$$

The conjugate pair $\{c, q - c\}$ is the minimal fibre of Π in the Weil setting, conditional on the absence of additional symmetries of S_{BI} beyond parity.

Proof. By Theorem 2.7, every fibre of Π contains the orbit $\{\chi, -\chi\}$. The identification $\chi \leftrightarrow c$ and $-\chi \leftrightarrow q - c$ follows from the complex conjugation $\rho_{q-c} = \overline{\rho_c}$, which is the spectral realisation of $\chi \mapsto -\chi$ in the Weil setting (Section 3.3). The minimality is inherited from Theorem 2.5 under the same conditional hypothesis. $\square$

4 Implications and relation to O16–O17

4.1 Requalification of O16 and O17

O16 [22] introduced the pair observable $\sigma_{\text{pair}}(n) = \sigma_c(n) \cdot \sigma_{q-c}(n)$ and stated the identification of $\{c, q - c\}$ with fibres of Π as a structurally motivated hypothesis (O16 Remark 6.2). O17 [23] derived the equality $\delta_{q-c} = \delta_c$ from $\rho_{q-c} = \overline{\rho_c}$ and identified $r(c, q)$ as a normalisation artefact, but left the fibre identification as the primary open problem (O17 Section 5.2). Theorem 3.3 closes this open problem: the identification is now a derived consequence of the Born–Infeld structure and is no longer a hypothesis.

Remark 4.1 (Complementarity with O15–O17). *O15 [21] established that no reweighting of block-level data can recover the fibre-level exponent. O16–O17 established the correct observable and exponent at the pair level. O18 (the present paper) provides the foundational derivation of the fibre structure underlying O16–O17. The three results are complementary: O15 closes the block-level programme; O16–O17 open and populate the fibre-level programme; O18 provides its structural foundation.*

4.2 Effect on $\delta_{\text{pair}} \rightarrow \beta^*$

The structural relation $\beta^* = 1/(\delta + 1/2)$, used throughout the O-series, should be evaluated at $\delta = \delta_{\text{pair}} \approx 7.44$ as established in O16–O17. This

evaluation was already performed in O17 Section 4 on the basis of the (then hypothetical) fibre identification. Theorem 3.3 removes the hypothetical status: the substitution $\delta \rightarrow \delta_{\text{pair}}$ is now a derived consequence of the fibre structure, not an assumption. This also removes the last remaining observable-level ambiguity identified in O15: the aggregation no-go of O15 Corollary 4.8 [21] showed that no block-level reweighting recovers the correct exponent; O18 now explains why the fibre level is the structurally mandated one. The full derivation of the $\delta_{\text{pair}} \rightarrow \beta^*$ relation at the fibre level, without reliance on pipeline conventions, is the subject of O21.

4.3 Open directions

The present paper leaves three directions open.

Canonical normalisation. The pipeline normalisation factor $D(c, b_1, b_2, n)$ is currently implicit in the O12/O13 implementation. Defining a canonical normalisation independent of (b_1, b_2) would eliminate r entirely from the pair observable, producing a version of σ_{pair} that is genuinely independent of the measurement convention. This is a technical, not conceptual, task and is the subject of O19.

Upper bound of the phenomenological window. Numerical exploration (post-O16) shows that the spectrum of δ_{pair} has a hard lower bound near 7.4 and extends above 10.6. The mechanism selecting $[7.4, 10.6]$ as the physical sub-spectrum is not yet identified. It appears to be a dynamical persistence criterion rather than a spectral one, and is the subject of O20.

General classification of fibres. Theorem 2.5 is conditional on the absence of additional symmetries. A full classification of fibres in frameworks with additional gauge or topological structure would determine whether larger fibres are possible and, if so, their structure. This programme is deferred to a subsequent paper following O21.

5 Summary

1. The Born–Infeld action $S_{\text{BI}}[\chi]$ is even in χ (Lemma 2.1). This follows directly from the dependence of S_{BI} on $F^2 = (D\chi)^2$.
2. Evenness of S_{BI} implies that every projective fibre of Π contains the parity orbit $\{\chi, -\chi\}$ (Corollary 2.4). This uses the definition of BI-indiscernability (Definition 2.3) and does not require any group-theoretic input.
3. Under the assumption that parity is the only global symmetry of S_{BI} , the minimal fibre is the involution $\Pi(\chi) = \Pi(-\chi)$ (Theorems 2.5 and 2.7). This is a conditional result; see Remark 2.6.

4. In the Weil realisation of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, the abstract involution is identified with $c \leftrightarrow q - c$ via the conjugation $\rho_{q-c} = \overline{\rho_c}$ established in O17 [23] (Theorem 3.3).
5. These results close the primary open problem stated in O17 Section 5.2. The pair observable $\sigma_{\text{pair}}(n) = \sigma_c(n) \cdot \sigma_{q-c}(n)$ and the exponent doubling $\delta_{\text{pair}} = 2\delta_c \approx 7.44$ are now structurally derived rather than hypothetically motivated.

References

- [1] J. Beau, “Cosmochrony: A Pre-Geometric Relational Framework for Emergent Physics,” white paper, Zenodo (2025). DOI: 10.5281/zenodo.17957509.
- [2] J. Beau, “Emergent Spacetime Geometry from Bounded Relational Relaxation,” companion paper, Zenodo (2026). DOI: 10.5281/zenodo.18407505.
- [3] J. Beau, “Spectral Admissibility under Bounded Flux: Non-Abelian Mode Selection on Cayley Graphs of $\text{SU}(2)$ Subgroups,” spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.18944985.
- [4] J. Beau, “Spectral Capacity Functional and the Binary-Polyhedral Maximality Conjecture,” spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.18945273.
- [5] J. Beau, “Three Particle Generations and Mass Hierarchy from Spectral Stratigraphy,” spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19040490.
- [6] J. Beau, “Asymptotic Saturation of Projective Resolution: Expander Relaxation Graphs,” spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19056975.
- [7] J. Beau, “Projective Resolution Dynamics on Variable-Valence Relaxation Graphs,” O1 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19076319.
- [8] J. Beau, “Projective Dynamics and Mass Hierarchy: Hierarchical Amplification via Growing Relational Valence,” O3 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19076244.
- [9] J. Beau, “Subdiffusive Valence Growth under Bounded Relational Flux: Structural Derivation of the Cascade Exponent β ,” O4 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19101472.

- [10] J. Beau, “Admissible Frontier Saturation and the Cascade Exponent: A Representation-Theoretic Obstruction and its Matrix-Level Refinement,” O5 spectr. adm. subprg. paper, Zenodo (2026). DOI: 0.5281/zenodo.19120328.
- [11] J. Beau, “Matrix-Level Dynamic Redundancy and the Structural Confinement of the Cascade Exponent β ,” O6 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19140624.
- [12] J. Beau, “Projective Capacity and the Continuum Limit of Admissible Redundancy,” O7 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19149773.
- [13] J. Beau, “Three-Step Path Fingerprints on LPS Graphs and the Capacity Growth Exponent,” O8 spectr. adm. subprg. paper, Zenodo (2025). DOI: 10.5281/zenodo.19151769.
- [14] J. Beau, “Projective Capacity Beyond Expanders: Polynomial-Growth Relaxation Graphs and the Capacity Exponent,” O9 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19154517
- [15] J. Beau, “Polynomial-Growth Capacity Dynamics at Large q : Volumetric Growth Confirmation, Capacity Decay, and the Weil Decomposition Roadmap,” O10 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19161074
- [16] J. Beau, “Weil-Block Projective Capacity on Heisenberg Graphs: Resolving the Representation Obstruction and Extracting the Capacity Exponent,” O11 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19169987
- [17] J. Beau, “Exact Weil-Block Projective Capacity on Heisenberg Graphs: Resolving the Final Obstruction to δ Extraction,” O12 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19194798.
- [18] J. Beau, “Asymptotic Stability of Exact Weil-Block Capacity on Heisenberg Graphs: Extended Prime Range, Variance Reduction, and Requalification of the $\delta \rightarrow \beta^*$ Tension,” O13 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19210831.
- [19] J. Beau, “Observable-Class Mismatch and the Corrected $\delta \mapsto \beta^*$ Relation on Heisenberg Graphs: Block Normalisation, Central-Phase Contribution, and Structural Correspondence,” O14 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19226272.
- [20] J. Beau, “Topological Invariants of Admissible Configurations: Charge Quantisation, Absence of Magnetic Monopoles, and Projective Chirality

- in Cosmochrony,” compaignon paper, Zenodo (2026). DOI: 10.5281/zenodo.19282962.
- [21] J. Beau, “Observable-Class Derivation Mismatch and the Exact Growth Equation: From Proxy Capacity to Block-Level Dynamics,” O15 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19281420.
- [22] J. Beau, “Fibre Admissibility and Resolution of the Capacity Exponent Discrepancy,” O16 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19298427.
- [23] J. Beau, “Fibre-Level Admissibility from Conjugate Weil Blocks: Structural Derivation of the Pair Observable,” O16 spectr. adm. subprg. paper, Zenodo (2026). DOI: 10.5281/zenodo.19299611